

## Part 3 of 5

# American Carrier Strike Groups: An Electronic Perspective



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The first two parts of this article, published in September and October issues, covered air defence warfare in great detail. In this part we discuss electronic warfare (EW), under-sea warfare, surface warfare and strike warfare.

## Electronic warfare

EW is denying the enemy the advantage of using the electromagnetic spectrum effectively. In detail, the objective is to degrade the performance of the enemy's communication and detection (radar) systems. This involves two operations, namely, detection and jamming/spoofing.

Detection is carried out through a scheme called electronic support measures (ESMs). This detects the enemy's radar activity or their communication transmissions.

Jamming/spoofing is carried out through electronic counter measures (ECMs). This degrades the performance of the enemy's radar activity and their communications. Through ESM, signals—be it radar or communication—are detected. If an imminent attack is indicated, those

signals are jammed through ECM.

Both ESM and ECM are integrated into a single module called EW module. Two such separate modules are fitted in the carrier, one on each side. These two are tied up to an EW command console to give an overall scenario of emissions around the carrier. Similarly, a set of EW modules is present in Aegis destroyers and cruisers.

**Electronic support measures.** There is an interesting and important property of radars. A radar is like a double-edged sword for the user. The ultimate intention of using it is to detect the enemy. How? A radar is like a flashlight used by an imaginary flycatcher to find a fly in a dark auditorium. The flycatcher has to find the fly by waving the flashlight around. There is an effective range for the flashlight within which the flycatcher can find the fly.

But what will happen if the fly is beyond the effective range of the flashlight? The fly can easily find the presence of the flycatcher by seeing the dimly-lit flashlight. The flycatcher cannot find the fly at that distance but the fly can find the presence of the flycatcher. For this, the fly must have sensitive eyes, and that is all.

Similarly, a radar has a range within which it can effectively detect/track the target. Beyond that range, electromagnetic waves do not stop but carry on their travel for a long distance while getting attenuated in due course. So to detect the presence of a radar, a simple high-power radio receiver, very much synonymous to the common radio receiver, tuned to the radar frequency suffices. ESM systems have many radio receivers scanning over different frequencies. If these detect any signal, these stop scanning and fine tune to that particular frequency. Once it is done, direction of that signal can be found out and, further, the type of system that emits it can be detected.

How is it done? Each type of radar has

Fig. 12: EW module in a destroyer

